Honours Multivariate Analysis
Continuous Assessment 6

Instructions:

- You will be divided into groups for this assessment. Only 1 submission per group is required.
- Your **.pdf** report may be compiled using any software you like (Rmarkdown, LaTeX, MSWord, etc.), as long as the presentation is neat.
- Do NOT paste R output verbatim, this will be penalised. If you want to include R output, typeset it properly or present it in a table.
- To help the reader easily assimilate the information, round values to a small number of decimal places (unless there is a reason for expressing a more exact value).

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1. Once again consider the Egyptian skull data from CA1, given in `CA1.csv`.
   (a) Use Box's M-test to test whether the covariance matrices for the five periods are equal, i.e. test $H_0\colon \mathbf{\Sigma}_1 = \mathbf{\Sigma}_2 = \cdots = \mathbf{\Sigma}_5$.
   (b) Construct a one-way MANOVA and test whether the mean vectors of all five time periods are equal. If there is a significant difference, is this the case for all four variables?

2. The data below, also provided in `CA6.txt`, show the amount of fabric wear (y_1, y_2, and y_3) in three successive periods: (1) the first 1000 revolutions, (2) the second 1000 revolutions, and (3) the third 1000 revolutions of an abrasive wheel. There were two factors: type of filler used, and proportion of filler. There were two replications.

	Proportion of Filler								
	P_1(25%)			P_2(50%)			P_3(75%)		
Filler	y_1	y_2	y_3	y_1	y_2	y_3	y_1	y_2	y_3
F_1	194	192	141	233	217	171	265	252	207
	208	188	165	241	222	201	269	283	191
F_2	239	127	90	224	123	79	243	117	100
	187	105	85	243	123	110	226	125	75

(a) Conduct a MANOVA using Pillai's trace as well as individual ANOVA's. What conclusions can you draw in terms of the effects of the factors on fabric wear? Does this seem to change over time?

(b) Clearly explain what interaction means in context of this example.

(c) Suppose the factory manager is specifically interested in the wear caused by the final 1000 revolutions. What advice would you provide, assuming the goal is to minimise the wear on the fabric?
